

SUMMARY

This video explains the foundational principles of Set Theory within the context of Discrete Mathematics for Computer Science. It clarifies formal definitions, cardinality, and the distinction between membership and inclusion. The document further explains power sets, infinite set equivalence via bijective functions, operational mechanics of union and intersection, and addresses misconceptions about the empty set and natural numbers in CS.

I. FUNDAMENTAL DEFINITIONS AND PROPERTIES OF SETS

A set is an unordered collection of distinct elements.
Elements may be heterogeneous.

1.1 Cardinality and Types of Sets

Cardinality

The number of elements in a set.

Finite Set

Contains a countable, finite number of elements.

Infinite Set

Contains infinitely many elements.

Infinite cardinality is conceptually defined but not treated numerically like finite sets.

Natural Numbers in Computer Science

In CS, 0 is included in the set of natural numbers.

II. SET RELATIONSHIPS: EQUIVALENCE AND EQUALITY

2.1 Equivalent Sets

Two sets are equivalent if they have the same cardinality.

Finite Sets

Equivalent if $|A| = |B|$.

Infinite Sets

Equivalent if a bijective function exists between them.

Example

Set of non-negative multiples of 2 is equivalent to Natural Numbers.

Counter-example

Real numbers in $(0,1)$ are not equivalent to Natural Numbers.

2.2 Equal Sets

$A = B$ if and only if:

$A \subseteq B$ and $B \subseteq A$

All equal sets are equivalent.

Equivalent sets are not necessarily equal.

III. SUBSETS AND POWER SETS

3.1 Subsets and Proper Subsets

Subset ($\subseteq$)

$A \subseteq B$ if every element of A belongs to B .

Properties

Every set is a subset of itself.

Empty set is a subset of every set.

Super Set

If $A \subseteq B$, then $B \supseteq A$.

Proper Subset

All subsets except the set itself.

3.2 Power Set $P(A)$

Definition

Set of all subsets of A .

Cardinality

If $|A| = n$, then $|P(A)| = 2^n$.

Subsets of Size k

Number of subsets of size k :

nCk

Summation Properties

Total subsets:

$\sum nC_i$ from $i = 0$ to $n = 2^n$

At most k elements:

$\sum nC_i$ from $i = 0$ to k

At least k elements:

$\sum nC_i$ from $i = k$ to n

IV. MEMBERSHIP VS INCLUSION

Distinction between $\in$ and $\subseteq$ is critical.

Operation	Symbol	Criteria for Truth
Membership	$\in$	LHS is a direct element of RHS
Inclusion	$\subseteq$	LHS is a set and all its elements are in RHS

Case Study: Empty Set and Power Set

For any set A:

$\emptyset \in P(A) \rightarrow \text{True}$

$\emptyset \subseteq P(A) \rightarrow \text{True}$

$\emptyset \in A \rightarrow \text{False (unless explicitly defined)}$

V. SET OPERATIONS

5.1 Primary Operations

Union ($A \cup B$)

Elements in A or B.

Commutative.

Intersection ($A \cap B$)

Elements common to both.

Commutative.

Disjoint Sets

$A \cap B = \emptyset$

Set Difference ($A - B$)

Elements in A but not in B.

Non-commutative.

$A - B = B - A$ only if $A = B$.

Complement (A^c)

Elements in Universal Set U but not in A.

5.2 Symmetric Difference (Δ)

Definition

$$A \Delta B = (A - B) \cup (B - A)$$

$$A \Delta B = (A \cup B) - (A \cap B)$$

Property

Commutative.

VI. MATHEMATICAL LAWS OF SET OPERATIONS

Idempotent Law

$$A \cup A = A$$

$$A \cap A = A$$

Identity Law

$$A \cup \emptyset = A$$

$$A \cap U = A$$

Domination Law

$$A \cup U = U$$

$$A \cap \emptyset = \emptyset$$